

Where Next? Identifying U.S. Metros at Risk of Housing Unaffordability

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Abstract & Motivation

This project develops an early-warning model to identify currently affordable U.S. metros whose housing-market trajectories resemble Tampa, Austin, and Boise before major affordability collapse (*when Price-to-Income ratio is greater than 5.0*). Housing, income, and population indicators are joined by CBSA code to rank metros by affordability-risk resemblance.

The primary question being: **Which housing-market and socioeconomic indicators best predict a metro's movement toward affordability stress?** The model supports housing departments, planners, lenders, and real-estate stakeholders by providing earlier signals to monitor supply constraints, affordability trends, and potential displacement exposure. A *high score indicates a need for closer review, not certain displacement or a causal prediction.*



Background & Related Work

Zillow, FHFA, and Census data describe housing prices, price appreciation, income, population growth, and local housing conditions, but they do not typically identify metros at risk of emerging affordability stress. This project integrates these sources at the CBSA level and uses Tampa, Austin, and Boise as historical pre-collapse examples. It evaluates whether similar patterns in housing costs, income, and population growth can serve as early warning signals in other currently affordable metros.



Methods & Evaluation

Approach: An XGBoost classifier is trained using five-year price-to-income change, three-year home-price growth trend, prior-year home-price growth, and prior-quarter home-value growth. Tampa, Austin, and Boise serve as labeled affordability-collapse trajectories, and the trained model scores currently more affordable comparison metros by their similarity to those pre-collapse patterns.

Validation: Backtesting holds out the final two to three pre-collapse years for each training metro and tests whether the model would have identified elevated risk using only earlier observations. Performance is evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC; cross-validation is used for model tuning and performance assessment across metro-period observations.



Data, Engineering & Ethics

Sources & scale: Zillow ZHVI, FHFA HPI, Census ACS, and Census Population Estimates are joined by CBSA code into a metro-period panel for feature engineering, model training, and risk scoring.

Pipeline: Python and R ingest, clean, standardize, and merge source data. The workflow includes quality checks, feature engineering, XGBoost scoring, SHAP interpretation, and reproducible GitHub storage.

Ethics & responsible use: This project uses public, aggregate metro-level data and contains no personally identifiable information. Because risk score may influence investment or policy attention, results are intended to support monitoring and early planning, not deterministic forecast.

1. Collect Data	2. Join Metro Data	3. Quality Check	4. Final CSV
Gather House Value files, City Code files, and Census API pull	Joining all files through CBSA codes	Quality check for missingness, data loss, any CBSA code errors	Final CSV - 83,699 rows and 25 columns



Discussion, Limitations & Conclusions

Interpretation: Risk scores rank metros by similarity to the historical pre-collapse trajectories of Tampa, Austin, and Boise. *They are early-warning signals for closer monitoring, not predictions of inevitable affordability collapse or causal explanations.*

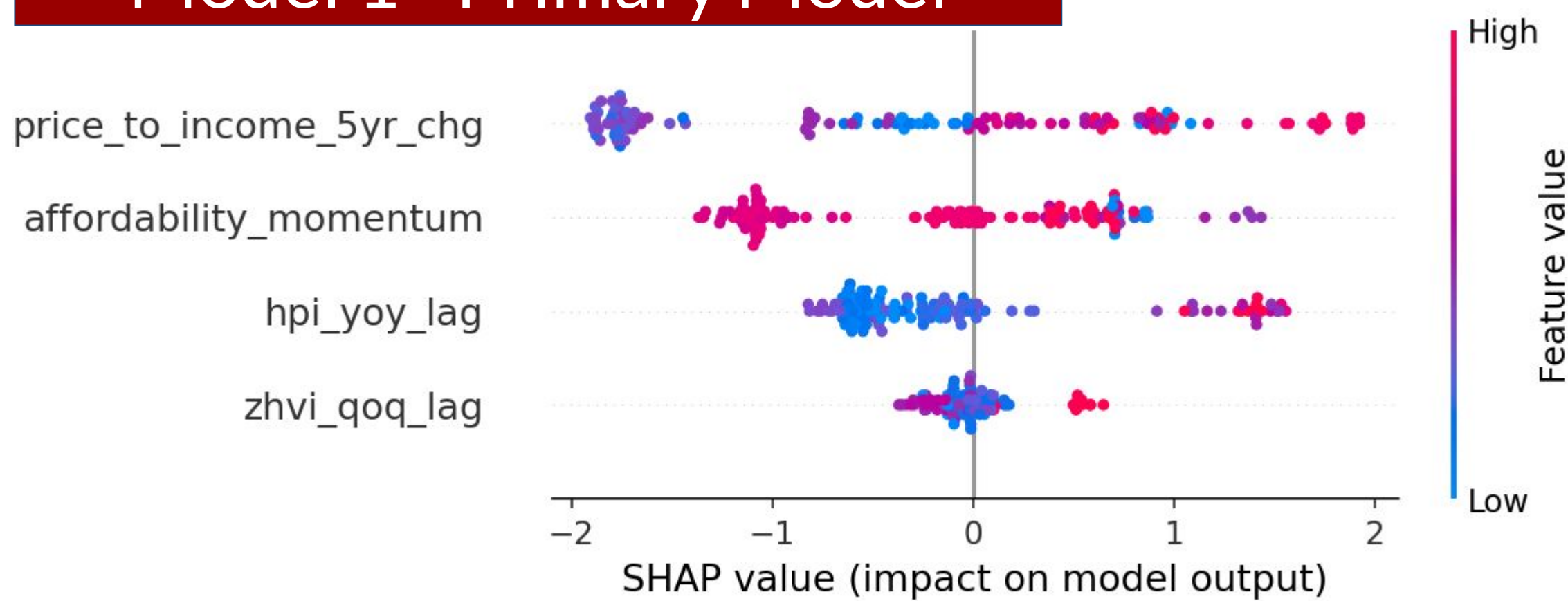
Limitations & future work: The model is limited by a small, geographically concentrated training set and lagged ACS data. It also excludes factors such as mortgage rates, inventory, permits and local supply-demand conditions. Because it is correlation-based, ranking may produce false positives or negatives and may generalize less well outside Sun Belt and Mountain West metros. Future work should add housing supply, rental, mortgage, and demand indicators to improve accuracy and generalizability.

Conclusion: Model 1 identified worsening price-to-income trends, affordability momentum, and prior-year house-price appreciation as the strongest affordability-risk indicators. Bozeman, Pinehurst-Southern Pines, and Amherst Town-Northampton most closely matched these pre-collapse patterns. Ranking support early monitoring, not a causal claim or deterministic forecast.



Results & Visualizations

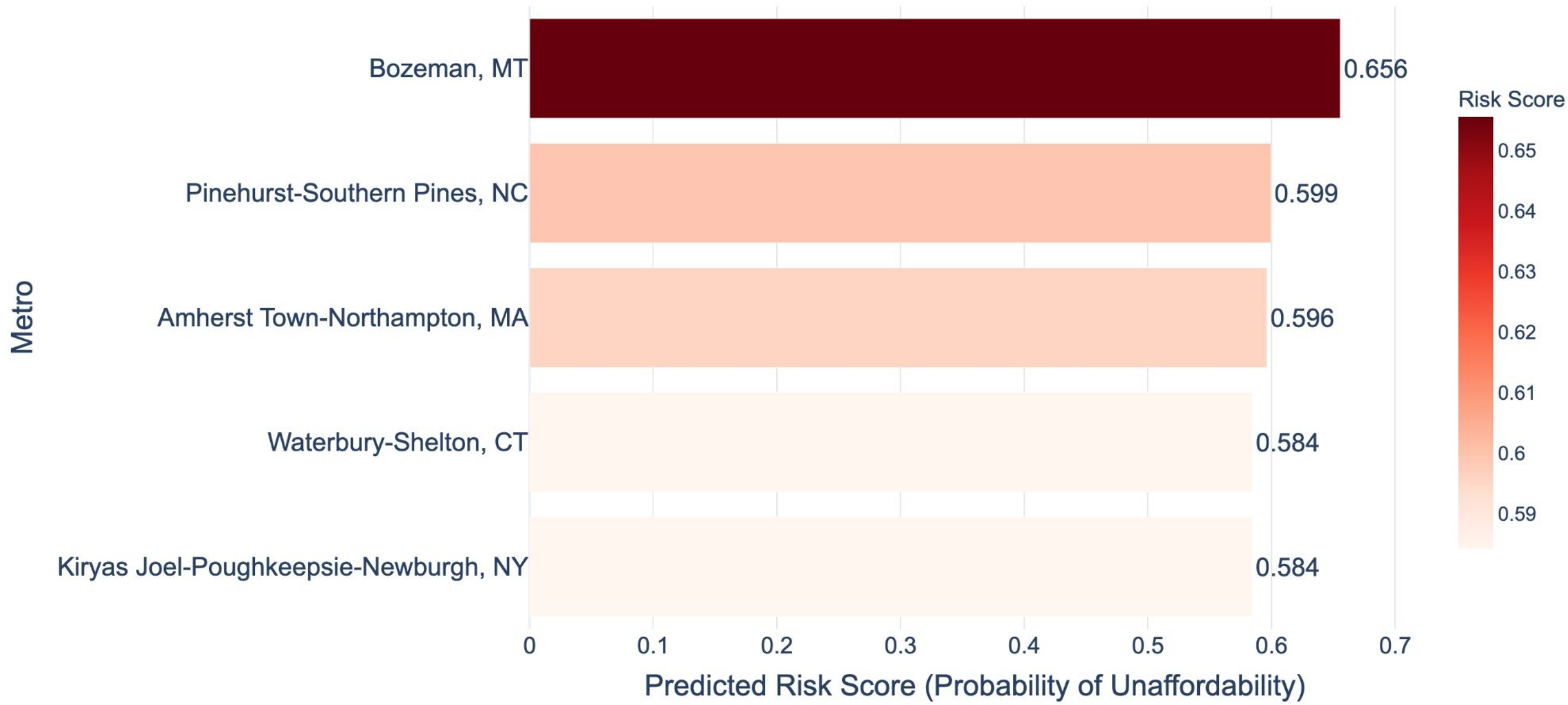
Model 1 - Primary Model



Feature	Average SHAP Importance
Five-Year Home Price-to-Income Change	1.083
Three-Year Home Price Growth Trend	0.702
Prior-Year Home Price Growth	0.588
Prior-Quarter Home Value Growth	0.138

Model 2 - Generalized Model

Top 5 Holdout Metros by Predicted Affordability Risk Score
Higher scores indicate stronger resemblance to pre-collapse Tampa, Austin, and Boise patterns



Metric	Score
Accuracy	78.8%
Precision	71.4%
Recall	50.0%
F1-Score	58.8%
ROC-AUC	0.713



Reproducibility & References

Repository: [<https://github.com/BSmith-cpu/DATA-510-CAPSTONE->] · Key dependencies: Python 3.x, pandas, scikit-learn, XGBoost, SHAP
References: Zillow Research ZHVI methodology; FHFA House Price Index documentation, Urban Institute housing cost-burden reports

